BUSINESS + TECHNICAL OVERVIEW

ProcureFlow AI

One connected workflow for purchasing, approvals and policy assistance.

Go • gRPC • Vue.js • PostgreSQL • RAG

PRESENTED BY

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Solution concept | November 2026



The idea in one sentence

A simple purchasing journey for people, backed by clear service boundaries for engineers.

Request → Review → Approve → Order → Track

Submit

Employees request goods or services with a reason and estimated cost.

Decide

Managers approve or reject with a recorded decision.

See progress

Procurement and employees follow each request from start to finish.

AI add-on: ask procurement policy questions and receive cited answers without leaving the application.

The problem it solves

Common procurement friction — illustrated as a general business scenario.

WITHOUT A SHARED SYSTEM

Scattered requests

Email threads and spreadsheets are difficult to follow.

Unclear status

Employees repeatedly ask whether a request is approved.

Hard-to-find policies

Approval rules live across separate documents.

WITH PROCUREFLOW AI

Structured requests

One form records what is needed, by whom and why.

Visible progress

Status and decision history are available in one view.

Searchable guidance

A policy assistant finds cited answers in approved documents.

These are typical pain points the concept targets, not measured results from a real organization.

Three people, one coordinated process

Each role has a focused set of actions and visibility.

Employee

Requests a purchase

- Create a purchase request
- Add quantity, cost and reason
- Track the status and decision

Manager

Makes the approval decision

- Review requests in a queue
- Approve or reject with reason
- See prior decisions / audit history

Procurement team

Processes approved orders

- View approved requests
- Record order and receipt status
- Monitor outstanding purchase work

AI assistant supports every role with document-backed policy guidance; it does not approve purchases.

A purchase request, end to end

Illustrative example: five laptops, €800 each → €4,000 total.

DEMO SCENARIO

1 Submit

Employee requests 5 laptops and adds business reason.

2 Validate

Purchase Service checks input and records the request.

3 Review

Approval Service creates a task for the manager.

4 Decide

Manager approves or rejects the request.

REJECTED

The reason is recorded; the employee sees the decision.

APPROVED

Procurement records the order, then marks it received.

Every state change can be displayed in a request timeline.

What users will actually see

Three simple screens make the workflow understandable and demo-friendly.

01 CREATE REQUEST

New purchase request

Item	Laptop
Quantity	5
Unit price	€800
Reason	Team equipment

SUBMIT REQUEST

02 MANAGER REVIEW

Pending approvals

PR-1024 • Laptop request

5 items | Total €4,000

PENDING

APPROVE REJECT

03 TRACK PROGRESS

Request PR-1024

- **Submitted**
- **Manager approved**
- Order placed
- Received

Illustrative interface mockups — not screenshots from an implemented application.

What makes it “AI”?

Two question types require two different sources of truth.

Policy questions

“Do purchases above €1,000 require approval?”

RAG looks up relevant policy passages → an LLM generates a grounded answer.

CITES SOURCE DOCUMENT

Live order questions

“What is the status of PR-1024?”

The assistant checks authorization, then asks the Go Purchase Service for the latest status.

USES CURRENT SYSTEM DATA

Read-only AI: no automatic purchasing decisions, approvals or supplier payments.

How we would judge success

Measure clarity and reliability - do not invent impact statistics.

Approval turnaround

Time from submission to a manager decision.

Request aging

How many requests are waiting beyond a target window.

Workflow visibility

Share of requests with a complete, current status.

Audit completeness

Share of decisions with actor, timestamp and reason.

Project goal: prove the workflow end to end, not claim real-world productivity gains before testing.

How the technology fits together

Each component has one clear job.

Vue.js

The screens people use

Forms, dashboards and a chat interface.

Go API Gateway

The front door

Accepts REST requests and routes them safely.

Go + gRPC services

The business engine

Purchase and approval rules communicate internally.

PostgreSQL

The system of record

Stores requests, decisions and supplier information.

RAG + pgvector

The policy knowledge layer

Retrieves document passages for grounded AI answers.

Docker Compose

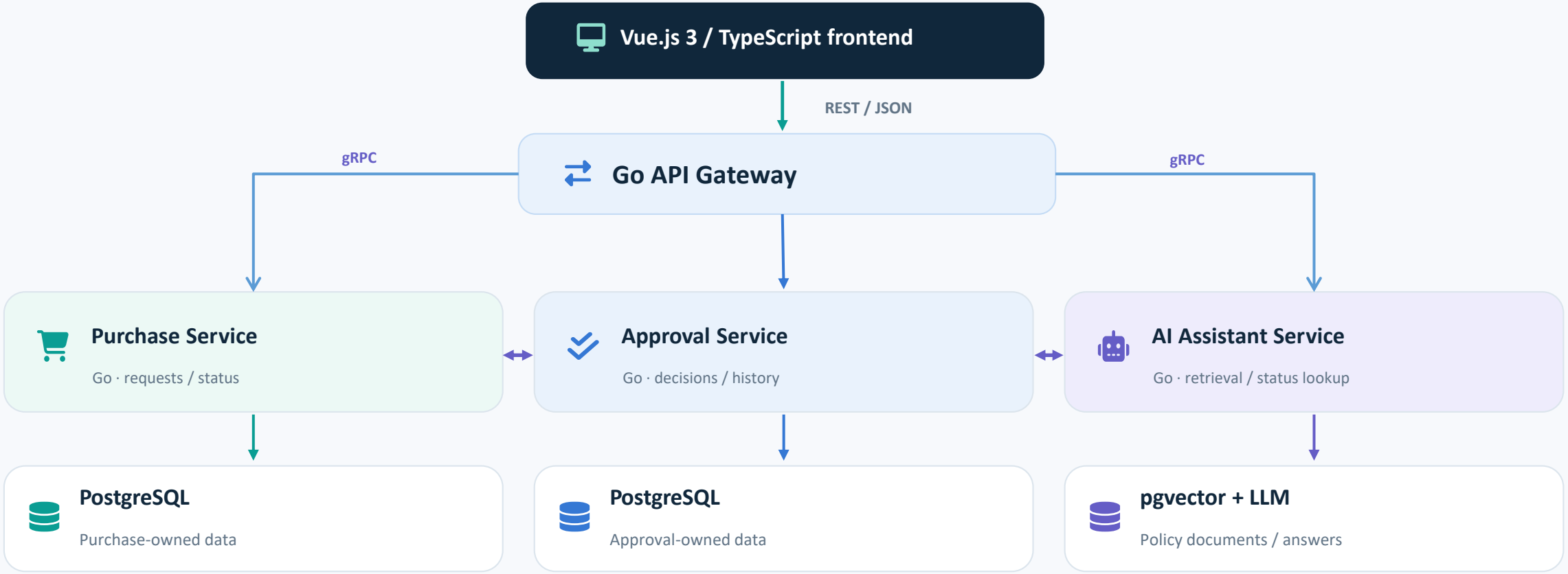
The local environment

Starts the services and dependencies together.

Business communication uses familiar terms; technical slides follow for engineers.

Technical architecture

Vue uses REST/JSON; Go services communicate through gRPC and Protocol Buffers.

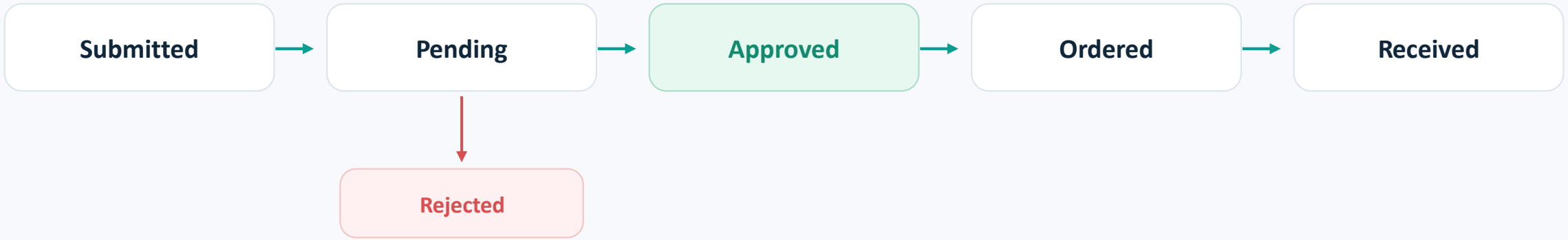


Conceptual boundaries: service-owned schemas may share one PostgreSQL instance in local Docker Compose.

Business rules become software rules

A clear state machine prevents invalid purchasing actions.

REQUEST LIFECYCLE



GUARDRAILS

Approval is human

Only an authorized manager decides.

Trace every decision

Actor, timestamp and reason are recorded.

No order before approval

Only approved requests can progress to ordered.

Built with reliability and trust in mind

Controls to design and verify - not claims of completed production hardening.

Access control

- Enforce roles server-side
- Protect requests by ownership

Correctness

- Validate totals and transitions
- Handle duplicate submissions safely

Service resilience

- Use gRPC deadlines / timeouts
- Log failures and recover safely

AI guardrails

- Use approved policy sources
- Keep AI read-only and cited

Demo uses fictional data; no live supplier payments, unattended approvals, or confidential policies.

Build the useful core first

One-day Codex sprint: prioritize a complete flow, then add AI only if stable.

1

Foundation

Repo, Go modules, .proto contracts

2

Business services

Purchase and approval gRPC services

3

Working application

PostgreSQL, REST gateway, Vue UI

4

Proof & packaging

Docker, tests, README, live demo

REQUIRED MVP

Create → approve/reject → track, with persisted data and tested gRPC calls.

STRETCH GOAL

RAG policy chatbot with citations.

Single Git branch: main. Small reviewed commits after each validated milestone.

PROCUREFLOW AI

A clear journey for buyers.

A meaningful technical demo for engineers.

5-MINUTE WALKTHROUGH

1. Submit a laptop request
2. Approve it as a manager
3. Show the updated status
4. Explain the gRPC service call
5. Optional: ask a cited AI policy question

Business value + service architecture + safe, explainable AI